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Centre Number

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Student Number

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION
(BOSS version)



Mathematics Extension 2

General

Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page

Total marks:

100

Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7–14)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Format Note:

Follows the 2026 exam format and 2017 NESA syllabus (not 2024).

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Consider the line segments in 3D space given by the vector equations below.

$$\vec{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, 0 \leq \lambda \leq 3; \quad \vec{s} = \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -4 \\ 2 \\ -4 \end{pmatrix}, -1 \leq \mu \leq 0.5; \quad \vec{v} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} + \gamma \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix}, 0 \leq \gamma \leq 2$$

Which of the following statements about the line segments is NOT true?

- A. At least two of them are parallel
- B. At least two of them are perpendicular
- C. At least two of them intersect
- D. At least two of them have the same length

2 Consider the statement:

$$\forall n \in \mathbb{Z}, n^3 \text{ is even.}$$

Which of the following is the negation of the statement?

- A. $\exists n \in \mathbb{Z}$, such that n^3 is odd.
- B. $\exists n \in \mathbb{Z}$, such that n^3 is even.
- C. $\forall n \in \mathbb{Z}, n^3$ is odd.
- D. n^3 is odd $\implies n \in \mathbb{Z}$

3 A complex number z is defined such that $|z - 3 - 4i| = 2$. Which of the following is the maximum value of $|z - i|$?

- A. $3\sqrt{2} - 2$
- B. $3\sqrt{2} + 2$
- C. 5
- D. 7

- 4 A particle is moving in simple harmonic motion along the x -axis. Its velocity $v \text{ ms}^{-1}$ at displacement x metres is given by the equation $v^2 = 9 + 16x - 4x^2$.

Which of the following correctly identifies the period of the motion and the maximum distance of the particle from the origin?

- A. Period π , Maximum distance $\frac{5}{2}$ m
- B. Period 2π , Maximum distance $\frac{5}{2}$ m
- C. Period π , Maximum distance $\frac{9}{2}$ m
- D. Period 2π , Maximum distance $\frac{9}{2}$ m
- 5 Consider the logical proposition P :
“For all real numbers x , there exists a real number y such that if $x > 0$, then $xy < 0$.”
- Which of the following represents the correct logical negation of proposition P ?
- A. There exists a real number x such that for all real numbers y , if $x > 0$, then $xy \geq 0$.
- B. For all real numbers x and for all real numbers y , $x \leq 0$ or $xy \geq 0$.
- C. There exists a real number x such that for all real numbers y , $x > 0$ **and** $xy \geq 0$.
- D. There exists a real number x such that there exists a real number y where $x > 0$ **and** $xy \geq 0$.
- 6 The complex numbers z_1 and z_2 satisfy the following conditions:

$$z_1 + z_2 = 6 + 8i$$

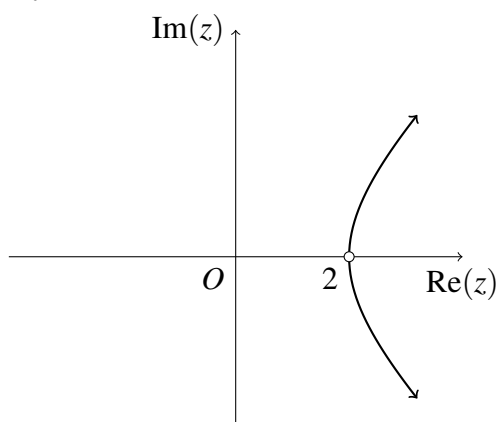
$$|z_1| + |z_2| = 14$$

What is the minimum value of $|z_1 - z_2|$?

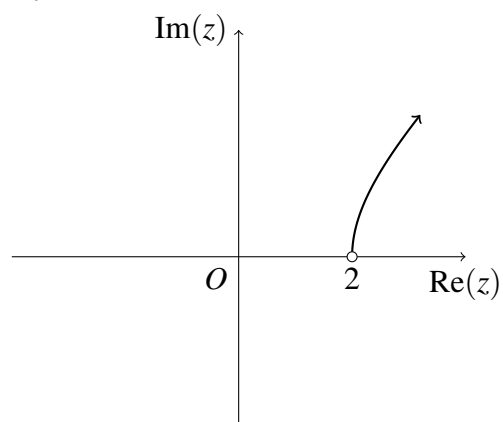
- A. $2\sqrt{6}$
- B. $4\sqrt{6}$
- C. 10
- D. 14

7 Which of the following Argand diagrams represents the locus of the complex number z that satisfies the equation $\text{Arg}(z-2) + \text{Arg}(z+2) = \frac{\pi}{2}$?

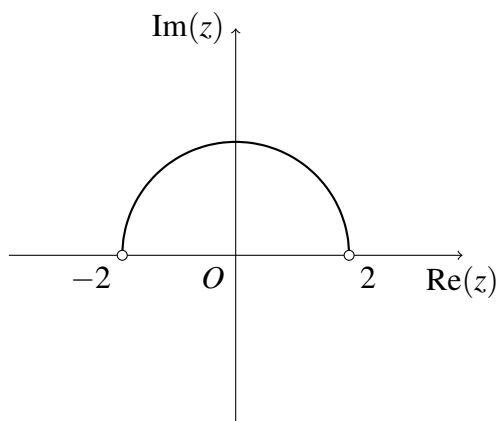
A.



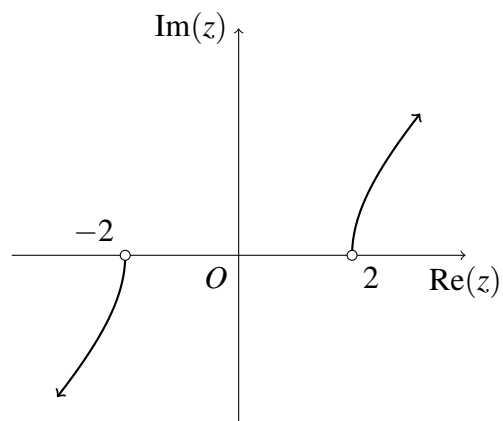
B.



C.

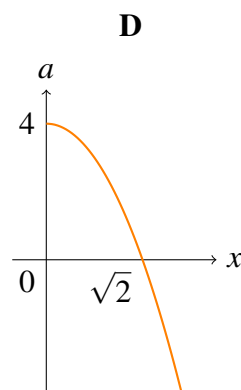
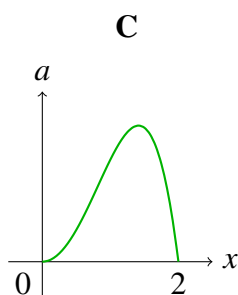
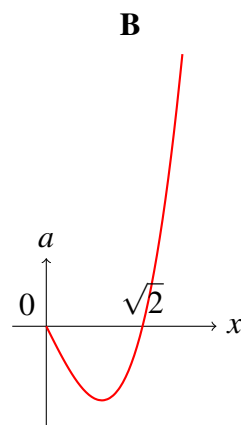
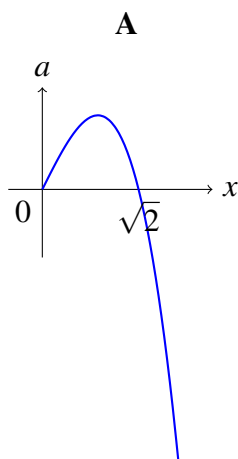


D.



- 8 A particle moves in a straight line such that its velocity v at displacement x is given by $v^2 = 4x^2 - x^4$, for $0 \leq x \leq 2$.

We want to represent the acceleration a of the particle as a function of its displacement x . Which colour do you like?



- 9 Let S be a sphere in 3-dimensional space defined by the vector equation $|\mathbf{r} - \mathbf{c}| = R$, where $\mathbf{c}$ is the position vector of its centre and R is its radius.

A line L is defined by the vector equation $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$, where $\mathbf{d}$ is a non-zero direction vector.

It is given that:

- A. $(\mathbf{a} - \mathbf{c}) \cdot \mathbf{d} = 0$
- B. $0 < |\mathbf{a} - \mathbf{c}| < R$

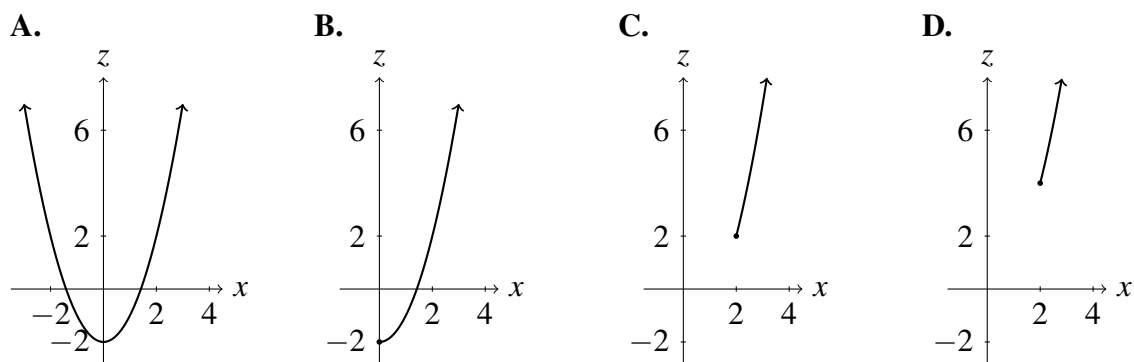
Which of the following best describes the geometric scenario between the sphere S , the line L , and the point $\mathbf{a}$?

- A. The point $\mathbf{a}$ is the midpoint of a chord formed by L intersecting S .
- B. The line L is tangent to the sphere S at the point $\mathbf{a}$.
- C. The point $\mathbf{a}$ is an endpoint of a chord formed by L intersecting S .
- D. The line L passes directly through the centre $\mathbf{c}$ of the sphere.

- 10 The curve C is defined by the following parametric equations for all real values of t :

$$x(t) = e^t + e^{-t}, \quad y(t) = \sin(t), \quad z(t) = e^{2t} + e^{-2t}$$

One of the following graphs represents the projection of the curve C onto the xz -plane. What character do you like the most. "A" for Awesome, "B" for Beautiful, "C" for Cute, "D" for Dazzling.



Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer the questions in the Section II Writing Booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (14 marks)

- (a) By considering the roots of $z^5 = 16\sqrt{2}(1 - i)$, show that: **2**

$$\cos \frac{\pi}{20} + \cos \frac{7\pi}{20} + \cos \frac{9\pi}{20} + \cos \frac{17\pi}{20} = \frac{\sqrt{2}}{2}$$

- (b) Let z be a non-zero complex number such that $\left|z - \frac{2i}{z}\right| = 1$. **2**
By using the triangle inequality, or otherwise, find the greatest possible value of $|z|$.

- (c) Let $z = 1 - i\sqrt{3}$ and $w = -1 + i$. **2**

Find the smallest positive integer n such that $(zw)^n$ is a purely imaginary number.

Question 11 continues on page 8

Question 11 (continued)

- (d) Let $\underline{p}$ and $\underline{q}$ be two non-zero 3-dimensional vectors. **3**

By considering the fact that $|\lambda \underline{p} + \underline{q}|^2 \geq 0$ for all real scalar values of λ , prove that:

$$(\underline{p} \cdot \underline{q})^2 \leq |\underline{p}|^2 |\underline{q}|^2$$

- (e) Let z be a complex number such that $|z| = 1$ and $z \neq -1$. **2**

Prove that $\frac{z-1}{z+1}$ is purely imaginary.

- (f) Find real constants A , B , and C such that, for all u , **3**

$$\frac{1}{u^3 + u^2 + u + 1} = \frac{A}{u+1} + \frac{Bu+C}{u^2+1}$$

Hence, evaluate

$$\int \frac{\sec^2 x}{\tan^3 x + \tan^2 x + \tan x + 1} dx$$

End of Question 11

Question 12 (16 marks)

- (a) Evaluate the following definite integral:

3

$$\int_0^{\frac{\pi}{4}} \frac{\sec^4 x}{(1 + \tan x)^3} dx$$

- (b) A sequence is defined recursively by
- $v_1 = \sqrt{3}$
- and for all integers
- $n \geq 1$
- :

3

$$v_{n+1} = \frac{v_n}{1 + \sqrt{1 + v_n^2}}$$

Prove by mathematical induction that for all integers $n \geq 1$,

$$v_n = \tan\left(\frac{\pi}{3 \cdot 2^{n-1}}\right)$$

- (c) On an Argand diagram, sketch the region representing the set of all
- z
- satisfying

3

$$1 \leq |z - 2i| \leq 3 \text{ and } \frac{\pi}{6} \leq \arg(z - 2i) \leq \frac{5\pi}{6},$$

showing the coordinates of any vertices in Cartesian form.

Hence, or otherwise, determine the maximum value of $|z|$ for any complex number z in this region.

- (d) (i) Express
- $P(x) = \frac{1}{x(1+x^2)^2}$
- in the fractional form
- $\frac{A}{x} + \frac{Bx+C}{1+x^2} + \frac{Dx+E}{(1+x^2)^2}$
- ,
-
- where
- A, B, C, D, E
- are constants.

2

$$(ii) \text{ Hence, find } \int \frac{1}{x(1+x^2)^2} dx.$$

2

- (e) Suppose a particle is moving horizontally along the
- x
- axis. Initially, the particle is at a displacement of
- $x = 2$
- metres from the origin and moving with a velocity of
- 3 ms^{-1}
- . The acceleration of the particle is given by
- $\ddot{x} = 4x - \frac{4}{x^3}$
- where
- $x > 0$
- .

$$(i) \text{ Show that the velocity of the particle is given by } \dot{x} = 2 \left(x - \frac{1}{x} \right).$$

1

$$(ii) \text{ Hence, find an expression for } x^2 \text{ as a function of } t.$$

2**End of Question 12**

Question 13 (15 marks)

- (a) Let
- $a > 0$
- . Evaluate the integral:

3

$$\int_0^a \frac{\ln(1+ax)}{1+x^2} dx$$

(You may use the property that $\int_0^c f(u) du = \int_0^c f(c-u) du$ without proof).

- (b) The points A , B , and C have position vectors $\tilde{a}$, $\tilde{b}$, and $\tilde{c}$ respectively, relative to an origin O . These four points form the vertices of a tetrahedron $OABC$.

- (i) The point P is the centroid (the point where the medians of the triangle intersect) of the face ABC . Let M be the midpoint of AB . Given that the centroid P divides the median CM internally in the ratio $2 : 1$, show that the position vector of P is given by

2

$$\tilde{p} = \frac{\tilde{a} + \tilde{b} + \tilde{c}}{3}$$

- (ii) The point Q is the centroid of the face OBC . Prove that the line segment PQ is parallel to the edge OA , and find the ratio of the length of PQ to the length of OA .

2

- (c) (i) Let $f(x)$ and $g(x)$ be continuous real-valued functions on the interval $[a, b]$. By considering $\int_a^b (tf(x) + g(x))^2 dx \geq 0$ for all real values of t , prove the Cauchy-Schwarz inequality for definite integrals:

2

$$\left(\int_a^b f(x)g(x) dx \right)^2 \leq \left(\int_a^b (f(x))^2 dx \right) \left(\int_a^b (g(x))^2 dx \right)$$

- (ii) Let $h(x)$ be a continuously differentiable real-valued function on the interval $[0, 1]$ such that $h(1) = 0$. By considering the derivative of $x(h(x))^2$, show that:

1

$$\int_0^1 (h(x))^2 dx = -2 \int_0^1 xh(x)h'(x) dx$$

- (iii) Hence, use the inequality from part (i) to prove that:

1

$$\int_0^1 (h'(x))^2 dx \geq \frac{1}{4} \int_0^1 (h(x))^2 dx$$

Question 13 continues on page 11

Question 13 (continued)

(d) For integers $m > 0$ and $n > 0$, we define the function $B(m, n)$ by the definite integral:

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

(i) Use the substitution $x = \frac{1}{1+u}$ to show that **2**

$$B(m, n) = \int_0^\infty \frac{u^{n-1}}{(1+u)^{m+n}} du.$$

Note: An integral over an infinite interval is evaluated using a limit,
such that $\int_0^\infty f(u) du = \lim_{R \rightarrow \infty} \int_0^R f(u) du$.

(ii) Hence, or otherwise, evaluate **2**

$$\int_0^\infty \frac{u^2}{(1+u)^6} du.$$

End of Question 13

Question 14 (15 marks)

(a) Let $I_n = \int_0^1 \frac{1}{(x^2 + 1)^n} dx$ for integers $n \geq 1$.

(i) Show that $2nI_{n+1} = 2^{-n} + (2n - 1)I_n$. 2

(ii) Hence, evaluate $\int_0^1 \frac{x^2}{(x^2 + 1)^3} dx$. 2

(b) A particle moves along a straight line such that its acceleration $a \text{ ms}^{-2}$ is given by

$$a = -3x^2v^3$$

where $v \text{ ms}^{-1}$ is the velocity of the particle and x metres is its displacement from the origin O .

Initially, the particle is at a displacement of $x = 1$ with a velocity of $v = \frac{1}{2}$.

(i) By expressing acceleration as $v \frac{dv}{dx}$, 2

show that the velocity of the particle is given by $v = \frac{1}{x^3 + 1}$.

(ii) Hence, find the exact time taken for the particle to reach the position $x = 2$. 2

(c) Let $\triangle ABC$ be a triangle in the complex plane with vertices A, B, C represented by the distinct complex numbers a, b , and c respectively.

(i) Prove that $\triangle ABC$ is equilateral if and only if $a^2 + b^2 + c^2 = ab + bc + ca$. 2

(ii) The complex numbers a, b , and c are the roots of the cubic equation: 2

$$z^3 - 3pz^2 + 3qz - r = 0$$

where p, q , and r are complex constants.

Using the result from part (i), or otherwise, deduce that $\triangle ABC$ is equilateral if and only if $p^2 = q$.

Question 14 continues on page 13

Question 14 (continued)

- (d) A sequence of real numbers is defined by $x_1 = 1$ and the recurrence relation: **3**

$$x_{n+1} = x_n + \frac{1}{x_n^2} \quad \text{for integers } n \geq 1$$

Prove by mathematical induction, or otherwise, that $x_n > \sqrt[3]{3n}$ for all integers $n \geq 2$.

End of Question 14

Premium Content

Want to test yourself on Questions 15 and 16?

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End of paper

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REFERENCE SHEET

Mathematics Advanced

Mathematics Extension 1

Mathematics Extension 2

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a+b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\text{For } ax^3 + bx^2 + cx + d = 0:$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

Financial Mathematics

$$A = P(1+r)^n$$

Sequences and series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, r \neq 1$$

$$S = \frac{a}{1-r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \cos A = \frac{\text{adj}}{\text{hyp}}, \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

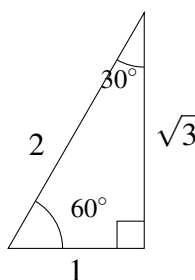
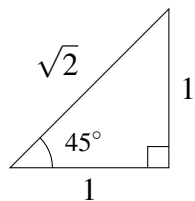
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$

**Trigonometric identities**

$$\sec A = \frac{1}{\cos A}, \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$$

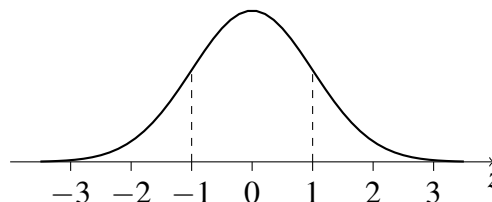
$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times \text{IQR}$
or
more than $Q_3 + 1.5 \times \text{IQR}$

Normal distribution

- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$
$y = uv$	$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$y = \frac{u}{v}$	$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\frac{dy}{dx} = f'(x) \cos f(x)$
$y = \cos f(x)$	$\frac{dy}{dx} = -f'(x) \sin f(x)$
$y = \tan f(x)$	$\frac{dy}{dx} = f'(x) \sec^2 f(x)$
$y = e^{f(x)}$	$\frac{dy}{dx} = f'(x) e^{f(x)}$
$y = \ln f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$
$y = \log_a f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$
$y = \sin^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$
$y = \cos^{-1} f(x)$	$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$
$y = \tan^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c \text{ where } n \neq -1$$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\} \text{ where } a = x_0 \text{ and } b = x_n$$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}||\underline{v}|\cos\theta = x_1x_2 + y_1y_2, \text{ where}$$

$$\underline{u} = x_1\underline{i} + y_1\underline{j} \text{ and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$z = a + ib = r(\cos\theta + i\sin\theta) = re^{i\theta}$$

$$[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta) = r^n e^{in\theta}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

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Centre Number

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Student Number

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION (BOSS version)

Mathematics Extension 2

Section II Writing Booklet

Instructions

- Use this writing booklet for Section II ONLY
- Write your Centre Number and Student Number at the top of this page
- Follow the instructions inside this booklet as to where you should write your answers
- Write the number of this booklet and the total number of booklets that you have used for this section (eg: **1** of **3**)

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 of

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this number of
booklet booklets for
 this section

BLANK PAGE

Do NOT write in this area.

Answer Question 11 on pages 3–6.

1	1
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Question Number

Do NOT write in this area.



Do NOT write in this area.



If you require more space to answer Question 11, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 11, tick here.

Do NOT write in this area.

End of Question 11

Go to page 8 for Question 12

1	2
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Question Number

Answer Question 12 on pages 8–11.

Do NOT write in this area.

Do NOT write in this area.



Do NOT write in this area.

If you require more space to answer Question 12, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 12, tick here.

End of Question 12

Go to page 13 for Question 13

Do NOT write in this area.

1 3

Question Number

Answer Question 13 on pages 13–16.*Do NOT write in this area.*



Do NOT write in this area.





If you require more space to answer Question 13, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 13, tick here.

Do NOT write in this area.

End of Question 13

Go to page 18 for Question 14

Answer Question 14 on pages 18–21.

1	4
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Question Number

Do NOT write in this area.

Do NOT write in this area.



Do NOT write in this area.

If you require more space to answer Question 14, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 14, tick here.

End of Question 14

Go to page 23 for Question 15

Do NOT write in this area.

1

5

Question Number

Answer Question 15 on pages 23–26.

Do NOT write in this area.



Do NOT write in this area.

Do NOT write in this area.

If you require more space to answer Question 15, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Question 15, tick here.

Do NOT write in this area.

End of Question 15

Go to page 28 for Question 16

1

6

Question Number

Answer Question 16 on pages 28–31.

Do NOT write in this area.

Do NOT write in this area.



If you have used an extra writing booklet for Question 16, tick here.

BLANK PAGE

Do NOT write in this area.